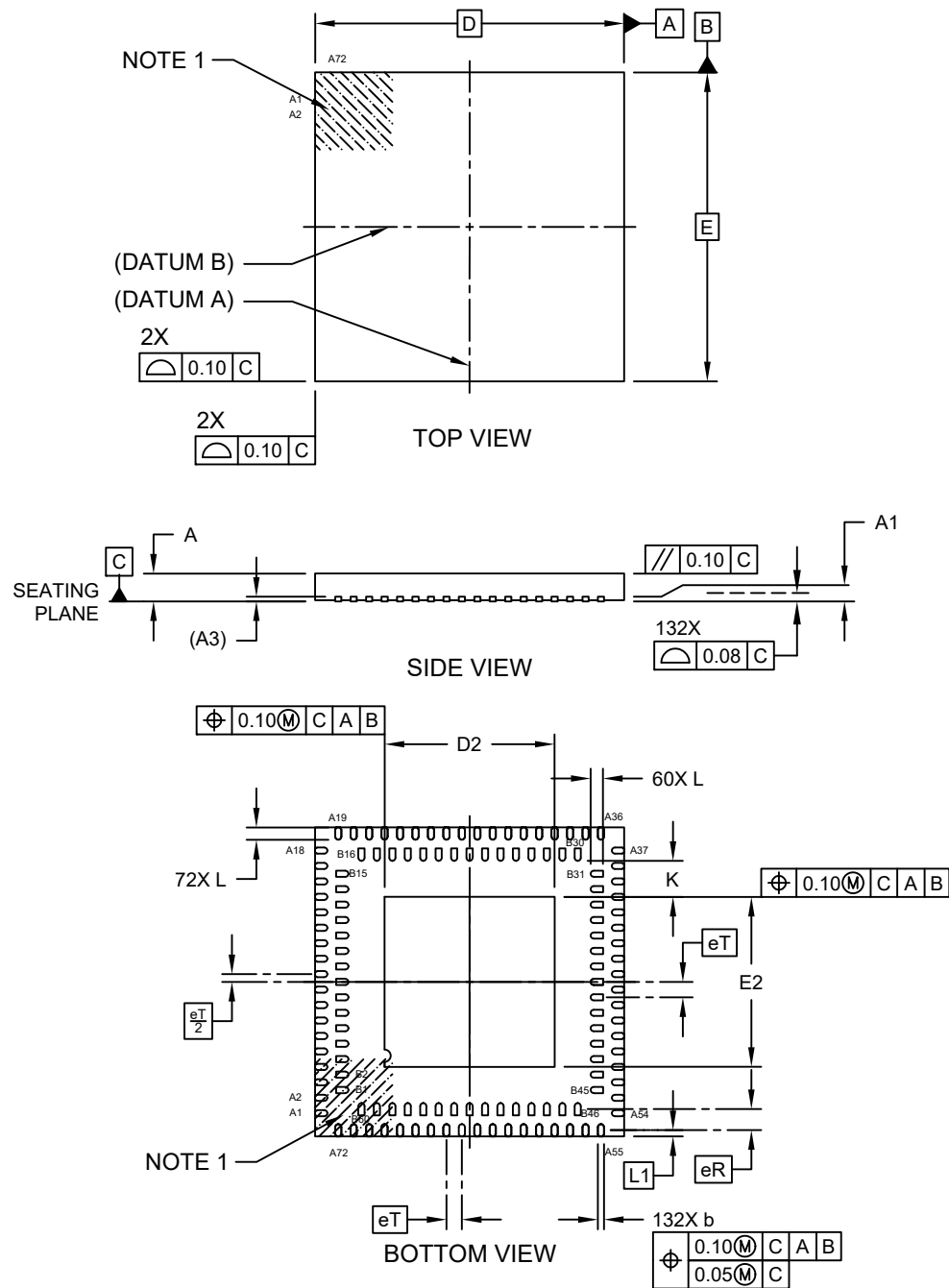


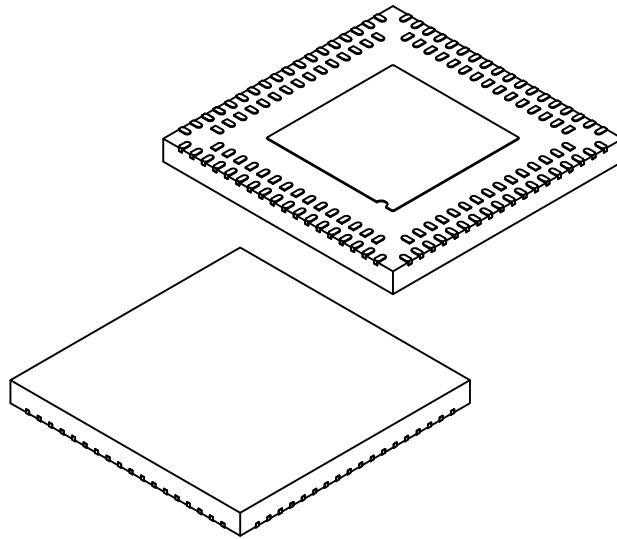
# 132-Lead Very Thin Plastic Quad Flat, No Lead Package (3WW) - 10x10x0.9 mm Body [VQFN] Dual Row Terminals; Saw Singulated

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



# 132-Lead Very Thin Plastic Quad Flat, No Lead Package (3WW) - 10x10x0.9 mm Body [VQFN] Dual Row Terminals; Saw Singulated

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



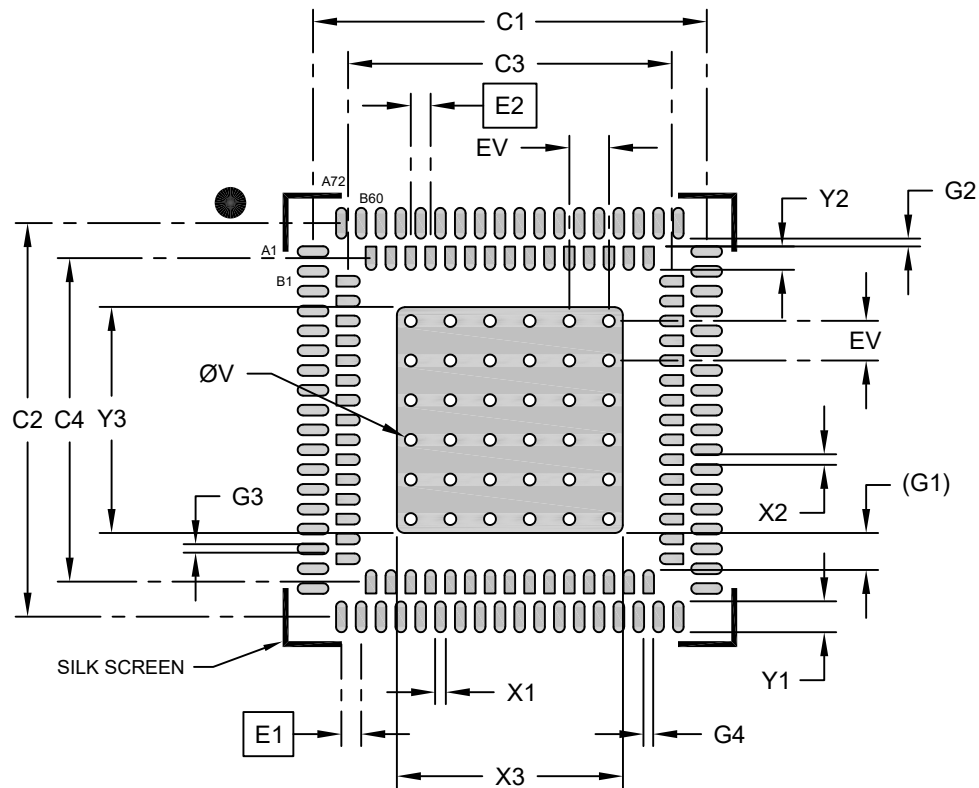
|                                      |    | Units | MILLIMETERS |      |      |
|--------------------------------------|----|-------|-------------|------|------|
| Dimension Limits                     |    |       | MIN         | NOM  | MAX  |
| Number of Terminals                  | N  |       | 132         |      |      |
| Pitch                                | eT |       | 0.50 BSC    |      |      |
| Outer Row Center to Inner Row Center | eR |       | 0.65 BSC    |      |      |
| Overall Height                       | A  |       | 0.80        | 0.85 | 0.90 |
| Standoff                             | A1 |       | 0.00        | 0.01 | 0.05 |
| Terminal Thickness                   | A3 |       | 0.152 REF   |      |      |
| Overall Length                       | D  |       | 10.00 BSC   |      |      |
| Exposed Pad Length                   | D2 |       | 5.40        | 5.50 | 5.60 |
| Overall Width                        | E  |       | 10.00 BSC   |      |      |
| Exposed Pad Width                    | E2 |       | 5.40        | 5.50 | 5.60 |
| Terminal Width                       | b  |       | 0.17        | 0.22 | 0.27 |
| Terminal Length                      | L  |       | 0.30        | 0.40 | 0.50 |
| Inner Row Terminal Length            | L2 |       | 0.30        | 0.40 | 0.50 |
| Terminal-to-Exposed-Pad              | K  |       | 0.20        | —    | —    |

## Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

**132-Lead Very Thin Plastic Quad Flat, No Lead Package (3WW) - 10x10x0.9 mm Body [VQFN]  
Dual Row Terminals; Saw Singulated**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**RECOMMENDED LAND PATTERN**

| Units                         |    | MILLIMETERS |      |      | Units                            |    | MILLIMETERS |      |      |
|-------------------------------|----|-------------|------|------|----------------------------------|----|-------------|------|------|
| Dimension Limits              |    | MIN         | NOM  | MAX  | Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Outer Contact Pitch           | E1 | 0.50 BSC    |      |      | Inner Contact Pad Spacing        | C4 |             | 8.16 |      |
| Inner Contact Pitch           | E2 | 0.50 BSC    |      |      | Outer Contact Pad Length (X72)   | Y1 |             |      | 0.78 |
| Outer Contact Pad Width (X72) | X1 |             |      | 0.30 | Inner Contact Pad Length (X60)   | Y2 |             |      | 0.59 |
| Inner Contact Pad Width (X60) | X2 |             |      | 0.30 | Contact Pad to Center Pad (X60)  | G1 | 1.20 REF    |      |      |
| Optional Center Pad Width     | X3 |             |      | 5.60 | Inner Pad Row to Outer Pad Row   | G2 | 0.20        |      |      |
| Optional Center Pad Length    | Y3 |             |      | 5.60 | Contact Pad to Contact Pad (X68) | G3 | 0.20        |      |      |
| Outer Contact Pad Spacing     | C1 |             | 9.93 |      | Contact Pad to Contact Pad (X56) | G4 | 0.20        |      |      |
| Outer Contact Pad Spacing     | C2 |             | 9.93 |      | Thermal Via Diameter             | V  |             | 0.30 |      |
| Inner Contact Pad Spacing     | C3 |             | 8.16 |      | Thermal Via Pitch                | EV |             | 1.00 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2575 Rev. C